Answer Key

Physics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
1	(1)	Vectors	Motion in a Plane
2	(4)	Real Gas	Kinetic Theory of Gases
3	(1)	Moment of Inertia	System of Particles and Rotational Motion
4	(2)	Image Formation by Lens	Ray Optics
5	(2)	Surface Charge Density	Electrostatics
6	(2)	River-Boat Problem	Motion in a Plane
7	(2)	Electric Field	Electrostatics
8	(3)	Electric Energy	Current Electricity
9	(1)	SHM	Oscillations
10	(1)	Magnetic Susceptibility and Permeability	Magnetism and Matter
11	(4)	Zener Diode	Semiconductor Electronics
12	(3)	Adiabatic Process	Thermodynamics
13	(3)	Force	Work, Energy and Power
14	(3)	Magnetic Field Intensity due to Curret Carrying Circular Coil	Moving Charges and Magnetism
15	(2)	Bohr's Atomic Model	Atoms
16	(2)	Moment of Inertia	System of Particles and Rotational Motion
17	(2)	Refraction at Curved Surfaces	Ray Optics
18	(2)	Dimensions	Units and Measurements
19	(4)	Applications of Gauss Law	Electrostatics
20	(3)	Photoelectric Equation	Dual Nature of Radiation and Matter
21	[150]	Pressure due to Liquid Column	Mechanical Properties of Fluids
22	[25]	Elastic Potential Energy	Mechanical Properties of Solids
23	[6]	YDSE	Wave Optics
24	[3]	Degree of Freedom	Kinetic Theory of Gases
25	[10]	Average Velocity	Motion in 1-D

Chemistry			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
26	(4)	Aromatic nature	General Organic Chemistry
27	(3)	Properties of alkyl halide	Halo alkane and arene
28	(3)	Group-17 properties	*p*-block
29	(4)	Henry's law	Liquid solution
30	(3)	Bohr radius	Structure of atom
31	(4)	Internal energy change	Thermodynamics
32	(4)	Raoult's law	Liquid solution
33	(3)	Concentration terms	Some basic concepts of chemistry
34	(3)	Condition of precipitate	Ionic equilibrium
35	(4)	Structure of molecule	Chemical bonding
36	(2)	Practical organic chemistry	Practical organic chemistry

37	(4)	Properties of amino acid	Biomolecules
38	(4)	Basic nature of amines	Amines
39	(2)	Group-13 properties	*p*-block
40	(4)	Crystal field theory	Coordination compound
41	(3)	Salt analysis	Practical inorganic chemistry
42	(1)	Empirical formula	Some basic concepts of chemistry
43	(4)	Homoletic cleavage	General Organic Chemistry
44	(4)	Rate of hydrolysis	Carboxylic acid
45	(1)	Shape of the molecule	Chemical bonding
46	[1]	Crystal field theory	Coordination compound
47	[6]	Nernst equation	Electrochemistry
48	[372]	Calculation of molecular weight	Some basic concept of chemistry
49	[74]	Equilibrium constant	Chemical equilibrium
50	[2435]	Half-life	Chemical kinetics

Mathematics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
51	(2)	Power of exponent	Permutation and combination
52	(3)	Latus rectum and eccentricity	Hyperbola
53	(3)	Number of sequence	Permutation and combination
54	(2)	Higher-order derivative	Differentiation
55	(4)	Product of matrix	Matrix
56	(1)	Independent term	Binomial thorem
57	(3)	Solution of trigonometric equation	Trigonometric equation
58	(1)	Sum of AP	Sequence and series
59	(4)	Maxima minima	Application of derivative
60	(1)	Distance from circle	Complex number
61	(3)	Projection	Vector
62	(2)	Types of relation	Set and relation
63	(1)	Limit with series expansion	Limit
64	(2)	System of equation	Determinant
65	(2)	Framing	Quadratic equation
66	(4)	Focal distance	Ellipse
67	(2)	Area of triangle	3D
68	(3)	Area of triangle	3D
69	(4)	Properties of adjoint	Matrix
70	(1)	Focal chord	Parabola
71	[8]	Properties of definite integration	Definite integration
72	[36]	Functional equation	Function
73	[22]	Area between curves	Applications of integration
74	[2477]	Probability	Probability
75	[768]	Circle equation	Circle

Answers with Explanation

Physics

1. Option (1) is correct.

The direction of propagation of a wave is normal to the wavefront $x + y + z =$ constant. The angle made by this direction with x-axis is given by

$$\cos\alpha = \frac{a_x}{\sqrt{a_x^2 + a_y^2 + a_z^2}} = \frac{1}{\sqrt{3}}$$

$$\alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

2. Option (4) is correct.

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

As per the rule of consistency of dimensions, every term joined by + or – should have same dimensions.

$$[P] = \left[\frac{a}{V^2}\right] \rightarrow [a]$$

$$= [PV^2] = [M^1L^5T^{-2}]$$

$$[V] = [b] = [L^3]$$

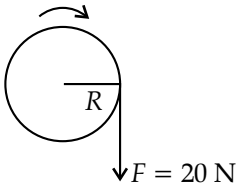
$$\therefore \quad [ab^{-2}] = \left[\frac{a}{b^2}\right] = \left[\frac{M^1L^5T^{-2}}{L^6}\right]$$

$$= [M^1L^{-1}T^{-2}]$$

$$\text{Energy density} = \frac{\text{Energy}}{\text{Volume}} = \mu$$

$$[\mu] = \left[\frac{M^1L^2T^{-2}}{L^3}\right] = [M^1L^{-1}T^{-2}]$$

3. Option (1) is correct.



Torque $\tau = FR = I_{CM}\alpha$

$$20R = MR^2\alpha$$

$$\alpha = \frac{20}{MR}$$

$$= \frac{20}{10 \times 0.1} = 20 \text{ rad/s}^2$$

$2\pi R$ metre $\rightarrow 2\pi$ rad

$$1 \text{ metre} \rightarrow \frac{2\pi}{2\pi R} \text{ rad}$$

$$= \frac{1}{R} \text{ rad} = \frac{1}{0.1} = 10 \text{ rad.}$$

$$\omega^2 = 2 \times \theta$$

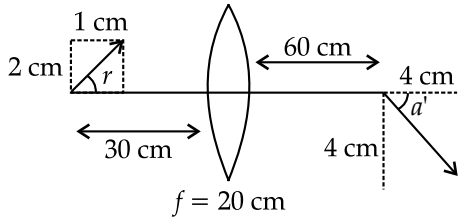
$$\omega = \sqrt{2 \times \theta}$$

$$= \sqrt{2 \times 20 \times 10}$$

$$= \sqrt{400}$$

$$= 20 \text{ rad/s}$$

4. Option (2) is correct.



Using Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$$v = 60 \text{ cm}$$

$$\text{Magnification } m = \frac{v}{u} = \frac{60}{-30}$$

$$= -2 = \frac{h_i}{h_o}$$

$h_i = -2 \times 2 = -4$ cm (inverted)

x-component of the slanted object has a length of 1 cm. Longitudinal magnification = $m^2 \times$ length of longitudinal object

$$= (2)^2 \times 1$$

$$= 4 \text{ cm}$$

From the diagram

$$\tan\alpha' = \frac{4}{4} = 1$$

$$\alpha' = 45°$$

Since, α' is measured in the clockwise direction, conventionally it is taken as –45°

5. Option (2) is correct.

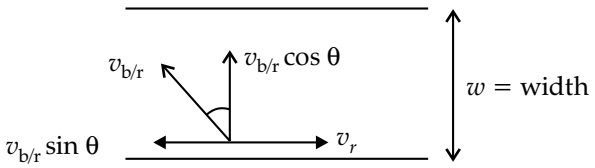
E_{Net} (at the midpoint) $= E_1 + E_2$

$$= \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} = \frac{3\sigma}{2\varepsilon_0}$$

Force F on point charge q is given by

$$qE_{Net} = \frac{3\sigma q}{2\varepsilon_0}$$

6. Option (2) is correct.



$v_r = 9$ km/h

$v_{br} = 27$ km/h

$\theta = 60°$

Time to cross the river, $t = \dfrac{w}{v_{br}\cos\theta}$

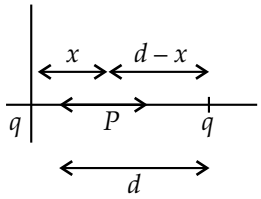
$w = v_{br}t\cos\theta$ (km)

$$= 27 \times \frac{30}{60 \times 60} \times \frac{1}{2} \text{ (km)}$$

$$= 27 \times \frac{30}{3600} \times \frac{1}{2} \times 1000 \text{ (m)}$$

$= 112.5$ m

7. Option (2) is correct.



Let P be a point at a distance x from the origin where $E = 0$

$$E_q = E_{9q}$$

$$\frac{kq}{x^2} = \frac{k(9q)}{(d-x)^2}$$

$$\frac{d-x}{x} = 3$$

$$d = 4x$$

$$x = \frac{d}{4}$$

8. Option (3) is correct.

Energy = Power × time

$= V \times I \times t$

$= 4.2 \times 5800 \times 10^{-3} \times 3600$ J

$= 4.2 \times 5800 \times 10^{-3} \times 3600 \times 10^{-3}$ (kJ)

$= 4.2 \times 5.8 \times 3.6$ (kJ)

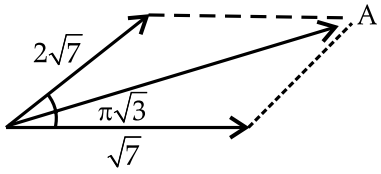
$= 87.7$ (kJ)

9. Option (1) is correct.

Given $x_1 = \sqrt{7}\sin 5t$ (cm)

$$x_2 = 2\sqrt{7}\sin\left(5t + \frac{\pi}{3}\right) \text{ (cm)}$$

Using phasors, we get



$$A = \sqrt{7 + 28 + 28\cos\frac{\pi}{3}}$$

$$= \sqrt{35 + 14}$$

$= 7$ cm

Maximum acceleration $a_{max} = \omega^2 A$

$= 25 \times 7 \times 10^{-2}$ m/s^2

$= 175 \times 10^{-2}$ m/s^2

$= x \times 10^{-2}$ m/s^2

On comparing, we get $x = 175$

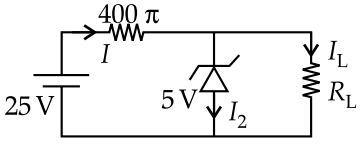
10. Option (1) is correct.

$1 + \chi = \mu_r$ (Relative permeability)

$$= \frac{\mu}{\mu_0}$$

$$\chi = \frac{\mu}{\mu_0} - 1$$

11. Option (4) is correct.



$V_{400} = 25 - 5 = 20$ V

$$I = \frac{V_{400}}{400} = \frac{20}{400}$$

$$= \frac{1}{20} \text{ A} = 50 \text{ mA}$$

$I_Z + I_L = 50$ (Junction Rule)

Given $I_Z = 4I_L$

$\therefore$ $5I_L = 50$

$I_L = 10$ mA

$$R_L = \frac{5}{10} = 0.5\ \text{k}\Omega$$

$$= 500\ \Omega$$

12. Option (3) is correct.

The molar heat capacity of a diabetic process is zero as $\Delta Q = nC\Delta T = 0$

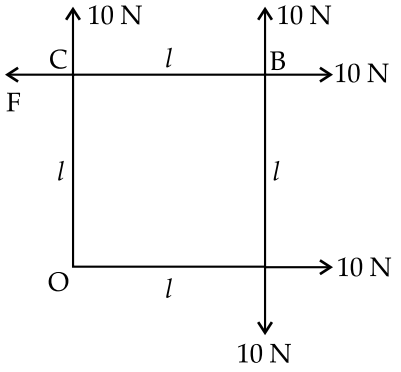
From the first law of thermodynamics

$\Delta Q = \Delta U + \Delta W = 0$

$\Delta W = -\Delta U$ = decrease in internal energy.

$\Delta U = nC_v\Delta T$. As T increases, U increases

13. Option (3) is correct.



Since, the square lamina is in equilibrium, the net torque about O is zero.

Now at point B, the resultant $F = 10\sqrt{2}$ N is along the diagonal passing through O. Hence, its torque is zero

Similarly, $\overrightarrow{F_1} = 10\hat{j}$ at C and $\overrightarrow{F_2} = 10\hat{i}$ at A pass through O, their torques about O are also zero.

Finally, $\overrightarrow{F_1}' = -F\hat{i}$ at C and $\overrightarrow{F_2}' = -10\hat{j}$ at A exert torques about O, which are mutually cancelled to keep the square lamina in equilibrium.

$10\ell = F\ell$

$\Rightarrow F = 10$ N

14. Option (3) is correct.

$$B_1 = \frac{\mu_0 I}{2R} \qquad \text{(at the centre)}$$

$$B_2 = \frac{\mu_0 IR^2}{2(R^2+x^2)^{\frac{3}{2}}}$$

(at an axial point)

Given $\frac{x}{R} = \frac{3}{4}$

$$x = \frac{3R}{4}$$

$$B_2 = \frac{\mu_o IR^2}{2R^3\left(1+\frac{9}{16}\right)^{\frac{3}{2}}}$$

$$= \frac{\mu_0 I}{2R}\left(\frac{16}{25}\right)^{\frac{3}{2}} = B_1\left(\frac{64}{125}\right)$$

$$\frac{B_2}{B_1} = \frac{64}{125}$$

15. Option (2) is correct.

$$E_n = -13.6\frac{Z^2}{n^2}(e\text{V})$$

For Option (A),

$$E_1 = -13.6\ (\text{eV})$$

$$E_2 = -13.6\times\frac{4}{4} = -13.6\ (\text{eV})$$

Option (A) is correct.

For Option (B)

$$E_1 = -13.6\ (\text{eV})$$

$$E_3 = -13.6\times\frac{9}{9} = -13.6\ (\text{eV})$$

Option (B) is correct.

For Option (C)

$$E_1 = -13.6\ (\text{eV})$$

$$E_1' = -13.6\times\frac{4}{1} = -54.4\ (\text{eV})$$

Option (C) is incorrect.

For Option (D)

$$E_2 = -13.6\times\frac{4}{4} = -13.6\ (\text{eV})$$

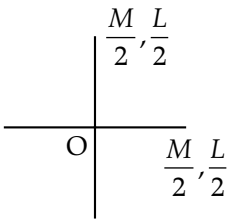
$$E_1 = -13.6\times\frac{9}{1} = -12.24\ (\text{eV})$$

Option (D) is incorrect.

16. Option (2) is correct.

Given $\alpha = \frac{ML^2}{12}$

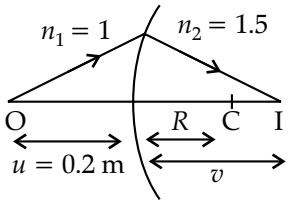
For O ——— M, L



$$I_0 = 2\times\frac{1}{12}\left(\frac{M}{2}\right)\left(\frac{L}{2}\right)^2$$

$$= \frac{ML^2}{48} = \frac{\alpha}{4}$$

17. Option (2) is correct.



Applying Gaussian formula

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{-0.2} = \frac{1.5-1}{0.4}$$

$$\frac{1.5}{v} + 5 = \frac{0.5}{0.4}$$

$$\frac{1.5}{v} = \frac{5}{4} - 5 = \frac{-15}{4}$$

$$v = \frac{-4\times1.5}{15} = -0.4 \text{ m}$$

–ve sign implies that the image is formed to the left of the spherical surface.

18. Option (2) is correct.

Coefficient of viscosity

$$[\eta] = \left[\frac{F}{A \times \left(\frac{dv}{dx}\right)}\right] = [M^1L^{-1}T^{-1}]$$

Intensity of wave

$$[I] = \left[\frac{E}{At}\right] = [M^1L^0T^{-3}]$$

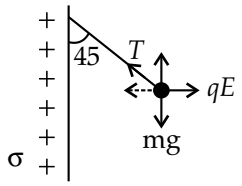
Pressure gradient

$$\left[\frac{p}{\ell}\right] = \left[\frac{F/A}{\ell}\right] = [M^1L^{-2}T^{-2}]$$

Compressibility

$$[C] = [M^{-1}L^{+1}T^{+2}]$$

19. Option (4) is correct.



At equilibrium, $\Sigma F_x = 0$ and $\Sigma F_y = 0$

$$T\cos45° = mg$$

$$T\sin45° = qE$$

$$\tan45° = \frac{qE}{mg} = 1$$

$$qE = mg$$

$$q\left(\frac{\sigma}{2\varepsilon_0}\right) = mg$$

$$\sigma = \frac{2\varepsilon_0 mg}{q}$$

Putting all values, we get

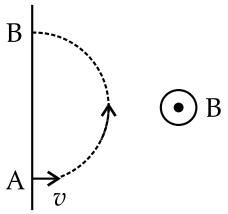
$$\sigma = \frac{2\times8.85\times10^{-12}\times10^{-4}\times10}{10^{-5}}$$

$$= 17.70 \times 10^{-10}$$

$$= 1.77 \times 10^{-9}$$

$$= 1.77 \text{ nC/m}^2$$

20. Option (3) is correct.



Since, $\vec{v} \perp \vec{B}$, emitted electron moves in a circular path and hits the plate at B.

$$AB = 2r = 2\left(\frac{mv}{qB}\right)$$

From Einstein's photoelectric equation

$$\frac{hc}{\lambda} = \phi + \frac{1}{2}mv^2 \text{ where}$$

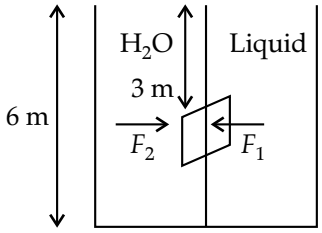
ϕ = work function

$$v = \sqrt{\frac{2}{m}\left(\frac{hc}{\lambda} - \phi\right)}$$

$$\therefore \quad AB = \frac{2m}{qB}\sqrt{\frac{2}{m}\left(\frac{hc}{\lambda} - \phi\right)}$$

$$= \frac{1}{qB}\sqrt{8m\left(\frac{hc}{\lambda}-\phi\right)}$$

21. Correct answer is [150].



$F_1 = (h_{\rho l} \cdot g)A + P_0A$

$F_2 = (h_{\rho w} \cdot g)A + P_0A$

$F_{Net} = F_1 - F_2$

$= h \times g \times A(\rho_e - \rho_w)$

$= 3 \times 10 \times 100 \times 10^{-4}\,(1.5 - 1) \times 10^3$

$= 150$ N (towards left)

External force to be applied to keep the hinged door closed = 150 N (towards right)

22. Correct answer is [25].

Poisson's ratio $\sigma = \dfrac{\Delta r/r}{\Delta \ell/\ell}$

$$\frac{\Delta \ell}{\ell} = \frac{\Delta r/r}{\sigma} = \frac{10^{-3}}{0.2}$$

$$= 5 \times 10^{-3} = \text{strain}$$

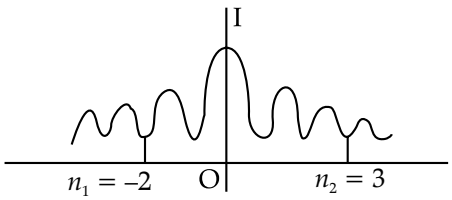
Elastic P.E. density

$$u = \frac{1}{2} \times Y \times (\text{strain})^2$$

$$= \frac{1}{2} \times 2 \times 10^{11} \times 25 \times 10^{-6}$$

$$= 25 \times 10^5 \text{ J/m}^3$$

23. Correct answer is [6].



For the above diffraction pattern, condition for minima is $a\sin\theta = n\lambda$,

$n = \pm1, \pm2, \pm3 \ldots$, a = slit width

For the second minima on the left, $a\sin\theta_2 = -2\lambda$

For the third minima on the right, $a\sin\theta_3 = 3\lambda$

$$\sin\theta_2 = \theta_2 = \frac{-2\lambda}{a}$$

$$\sin\theta_3 = \theta_3 = \frac{3\lambda}{a}$$

Given $\theta_3 - \theta_2 = \dfrac{5\lambda}{a} = \dfrac{\pi}{6}$

$$a = \frac{30\lambda}{\pi} = \frac{30 \times 628 \times 10^{-9}}{3.14} = 6\ \mu\text{m}$$

24. Correct answer is [3].

For a gas with f degrees of freedom, $\gamma = 1 + (2/f)$

$$\gamma_A = 1 + \frac{2}{3} = \frac{5}{3}$$

$$\gamma_B = 1 + \frac{2}{7} = \frac{9}{7}$$

$$\frac{\gamma_A}{\gamma_B} = \frac{5}{3} \times \frac{7}{9} = \frac{35}{27}$$

$$= 1 + \frac{1}{n}$$

$$\frac{1}{n} = \frac{8}{27} \Rightarrow n = \frac{27}{8} = 3.3 \approx 3$$

25. Correct answer is [10].

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time}}$$

$$\frac{50}{7} = \frac{x + \frac{3}{2}x}{t_1 + t_2} = \frac{\frac{5x}{2}}{\frac{x}{v_1} + \frac{3x}{2v_2}}$$

$$\frac{50}{7} = \frac{\frac{5}{2}}{\frac{1}{5} + \frac{3}{2v_2}}$$

$$\frac{35}{2} = 10 + \frac{75}{v_2}$$

$$\Rightarrow \quad \frac{15}{2} = \frac{75}{v_2}$$

$$\Rightarrow \quad v_2 = 10 \text{ m/s}$$

Chemistry

26. Option (4) is correct.

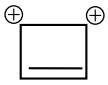
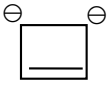
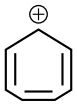
Those compounds are aromatic which

$\Rightarrow$ follow Huckel's rule of aromaticity $(4n + 2)\pi e^-$

$\Rightarrow$ are Cyclic

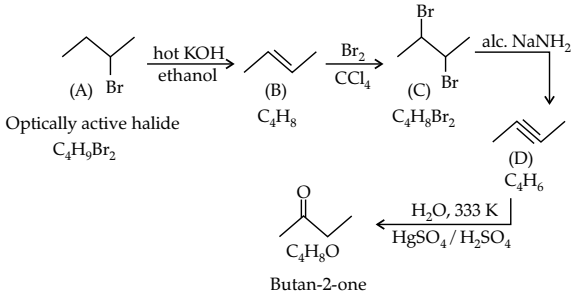
$\Rightarrow$ are Planar

$\Rightarrow$ have Complete conjugation

	$(4n+2)\ \pi e^-$	Cyclic	Planar	Complete conjugation	
(A) 	✓	✓	✓	✓	Aromatic
(B) 	×	✓	✓	✓	Not aromatic
(C) 	✓	✓	✓	✓	Aromatic
(D) 	✓	✓	✓	✓	Aromatic
(E) 	✓	✓	✓	✓	Aromatic
(F) 	×	✓	✓	✓	Not aromatic
(G) 	×	✓	✓	✓	Not aromatic
(H) 	✓	✓	✓	✓	Aromatic

(A), (C), (D), (E), (H) are aromatic and (B), (F), (G) are not aromatic

27. Option (3) is correct.



28. Option (3) is correct.

Group-17 belongs to halogen family, i.e., F, Cl, Br, I

(A)	Covalent radius	F < Cl < Br < I
(B)	e^-affinity	Cl < F < Br < I
(C)	Ionic radius	F < Cl < Br < I
(D)	First ionization energy	F < Cl < Br < I

The property which shows irregularity in the first four element of group-17 is e^- affinity.

The highest e^- affinity is of chlorine not fluorine.

29. Option (4) is correct.

The value of K_H depends upon temperature. The value of K_H increases with an increase of temperature implying that the solubility decreases with an increase at the same pressure.

Also it's value depends upon the nature of the gas and the solvent.

30. Option (3) is correct.

Formula used $r_n = a_0 \times \frac{n^2}{Z}$

where, a_0 = Bohr radius
n = orbit number
Z = atomic number

(1) When $n = 3$, $r_3 = a_0 \times \frac{3^2}{1} = 990$

Radius of third orbit is nine times larger than that of first orbit.

(2) When $n = 8$, $r_8 = a_0 \times \frac{(8)^2}{1} = 64\, a_0$

$$n = 4,\ r_4 = a_0 \times \frac{(4)^2}{1} = 1690$$

Radius of eighth orbit is four times larger than that of fourth orbit.

(3) When $n = 6$ $r_6 = a_0 \times \frac{(6)^2}{1} = 36\, a_0$

$$n = 4,\ r_4 = a_0 \times \frac{(4)^2}{1} = 16\, a_0$$

Radius of sixth orbit is four times larger than that of fourth orbit. It is incorrect statement.

(4) When $n = 4$, $r_4 = a_0 \times \frac{(4)^2}{1} = 16\, a_0$

$$n = 2,\ r_4 = a_0 \times \frac{(2)^2}{1} = 4\, a_0$$

Radius of fourth orbit is four times larger than that of second orbit.

31. Option (4) is correct.

After opening the stopcock, the gas from vessel A expands into vessel B and no change in the temperature is observed in the thermometer that indicates there is no pressure present in the vessel B due to which the gas readily expands from vessel A to B.

As $dT = 0 \therefore dU = 0$

$dU = 0 \therefore dq = 0$

From the first law of thermodynamics,

$dU = dq + dw$

So, $dw = 0$

$\therefore$ Only statement (4) is correct.

32. Option (4) is correct.

Given, $n_A = 1$ mol, $n_B = 3$ mol, $n_{total} = 4$

$P_A^\circ = 200$ mm Hg

$P_B^\circ = x$

$P_s = 500$ mm Hg

$$X_A = \frac{n_A}{n_{total}} = \frac{1}{4}\ X_B = \frac{n_B}{n_{total}} = \frac{3}{4}$$

From $P_s = P^\circ_A X_A + P_B{}^\circ X_B$ (Raoult's law)

$$500 = 200 \times \frac{1}{4} + x \times \frac{3}{4}$$

$500 = 50 + 0.75\, x$

$450 = 0.75\, x$

$$x = \frac{450}{0.75} = 600 \text{ mm Hg}$$

The vapour pressure of pure B = 600 mm Hg. The least volatile component of the sol^n = A, because its vapour pressure is less.

33. Option (3) is correct.

$$\text{No. of moles of } CaCO_3 = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{1000}{10} = 10 \text{ mol}$$

$$\text{mol of HCl} = M \times V_{(L)} = 0.76\ M \times 0.25\ L = 0.19 \text{ mol}$$

$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

Given 10 mol of $CaCO_3$ reacts with – 0.19 mol of $CaCO_3$.

According to stoichiometry 1 mol of $CaCO_3$ requires 2 mol HCl.

So 10 mol $CaCO_3$ requires 20 mL HCl

So, HCl is a limiting reagent.

Mass of $CaCl_2$ formed will be decided by HCl.

2 mol of HCl produces 111 g $CaCl_2$.

So, 0.19 mol of HCl produces $\frac{111 \times 0.19}{2} = 10.54$ g

34. Option (3) is correct.

$AY_2 \rightleftharpoons A^{2+} + 2Y^-$

Here, if $K_{ip} > K_{sp}$, then the precipitation of AY_2 takes place.

(1) $A^{2+} = 3.6 \times 10^{-3}$ M

$K_{ip} = [3.6 \times 10^{-3}][5 \times 10^{-4}]^2$

$[Y^-] = 5 \times 10^{-4}$ M shift

$K_{ip} = 9 \times 10^{-10}$

So, when $K_{ip} < K_{sp}$, no precipitate is formed.

(2) $[A^{2+}] = 2 \times 10^{-4}$ M

$k_{ip} = [2 \times 10^{-4}][0.8 \times 10^{-3}]^2$

$[Y^-] = 0.8 \times 10^{-3}$ M

$k_{ip} = 1.28 \times 10^{-10}$

So, when $k_{ip} < k_{sp}$, no precipitate is formed.

(3) $A^{2+} = 2 \times 10^{-2}$ M

$k_{ip} = [2 \times 10^{-2}][2 \times 10^{-2}]^2$

$[Y^-] = 2 \times 10^{-2}$ M

$k_{ip} = 8 \times 10^{-6}$ M

So, when $k_{ip} < k_{sp}$, precipitate is formed.

(4) $A^{2+} = 1.5 \times 10^{-4}$ M

$Y^- = 1.5 \times 10^{-3}$ M

$k_{ip} = [1.5 \times 10^{-4}][1.5 \times 10^{-3}]^2$

$= 3.375 \times 10^{-10}$ M

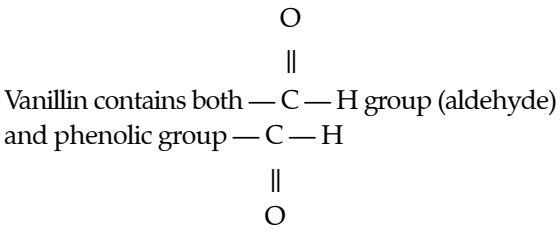
So, when $k_{ip} < k_{sp}$, no precipitate is formed.

35. Option (4) is correct.

	Hybridisation	No. of lone pairs	Dipole moment
SO_2	sp^2	1	Non-zero
NF_3	sp^3	1	Non-zero
NH_3	sp^3	1	Non-zero
XeF_2	sp^3d	3	Zero
ClF_3	sp^3d	2	Non-zero
SF_4	sp^3d	1	Non-zero

The hybridisation of the molecule with non-zero dipole moment & highest no. of lone pairs of e^- on the central atom is sp^3d.

36. Option (2) is correct.



Vanillin contains both — C — H group (aldehyde) and phenolic group — C — H

Aldehyde group gives positive Tollen's test while phenolic group reacts with NaOH.

For aldol reaction, α-H atom must be present but here aldehyde group does not contain α-H atom to give aldol condensation reaction.

37. Option (4) is correct.

(1) Is incorrect. Some naturally occuring amino acids contain more than one chiral centres.

(2) Is incorrect. Except glycine all naturally occuring amino acids are optically active.

(3) Apart from glutamic acid, aspartic acid also contains —COOH group at the side chain.

(4) Cysteine contains free —SH group due to which it undergoes dimerisation.

So, correct statement is (4).

38. Option (4) is correct.

The correct increasing order of basic nature is $NH_2—NH_3 < CH_3CH_2NH_2 < (CH_3CH_2)_3N < (CH_3CH_2)_2NH$

This is because the basicity of amines in an aqueous solution is governed by the following three effects:

(I) +I-effect of the alkyl group

(II) H-bonding with water molecules

(III) Steric effects

39. Option (2) is correct.

Statement I is incorrect.

- Metallic radius of Al (143 pm) is greater than that of Ga (135 pm).

Statement II is correct.

- The ionic radius of Al^{3+} (53.5 pm) is smaller than that of Ga^{3+} (62 pm).

40. Option (4) is correct.

Statement I is correct.

In octahedral complex, when the value of pairing energy (P) is higher than that of Δ_o, paring of e^- does not take place by ligand due to which high-spin complexes are formed.

When the value of pairing energy (P) is lower than that of Δ_o, pairing of e^- takes place by ligand due to which low spin complex are formed.

Statement II is correct.

In tetrahedral complex, the value of pairing energy is (P) high as compared to Δ_t. Therefore, pairing does not take place and we obtain low-spin complexes.

41. Option (3) is correct.

(A) The formation of chocolate brown precipitate takes place when $CuSO_4$ reacts with $K_4[Fe(CN)_6]$:

$$2Cu^{2+} + K_4[Fe(CN)_6] \rightarrow Cu[Fe(CN)_6] \downarrow + 4K^+$$

Chocolate brown ppt.

(B) The formation of prussian-blue precipitate takes place when $FeCl_3$ reacts with $K_4[Fe(CN)_6]$:

$$4Fe^{3+} + [Fe(CN)_6]^{4-} \rightarrow Fe_4[Fe(CN)_6]_3 \downarrow$$

Prussian blue

(C) Reaction of Zn^{2+} with $K_4[Fe(CN)_6]$ neutralised with NH_4OH gives white or bluish-white precipitate:

$$3Zn^{2+} + 2K^+ + 2[Fe(CN)_6]^{4-} \rightarrow K_2Zn_3[Fe(CN)_6]_2 \downarrow$$

White ppt.

42. Option (1) is correct.

Given, 1.46 g of CO_2 and 0.567 g of H_2O

$$n_{CO_2} = n_{Carbon} = \frac{1.46}{44} = 0.033 \text{ mol}$$

$$n_{H\text{-atom}} = 2 \times n_{H_2O} = \frac{0.567}{18} \times 2 = 0.063 \text{ mol}$$

Mole of C = 0.033 mol

Mass of C = 0.033 × 12 = 0.39 g

Mole of H = 0.063 mol

Mass of H = 0.063 g

Mass of O-atom = 1.0 – (0.39 + 0.063)

= 0.547 g

$$\text{Mole of O-atom} = \frac{0.547}{16} = 0.034 \text{ mol}$$

	Mole	Simple ratio	Empirical formula
C	0.033	$\frac{0.033}{0.033} = 1$	
H	0.063	$\frac{0.063}{0.033} = 2$	
O	0.034	$\frac{0.034}{0.033} = 1$	CH_2O

Empirical mass of CH_2O = 12 + 2 + 16 = 30 g.

43. Option (4) is correct.

The most stable free radical formed by carbon II due to resonance due to electron deficient *sp* hybridised C-atom and least stable free radical by I.

44. Option (4) is correct.

The rate of hydrolysis depends on the tendency of the leaving group.

Weak bases are good leaving group and order of leaving group are:

$$—Cl > —O—\overset{\overset{O}{\|}}{C}—CH_3 > —OCH_2CH_3 > —NH_2$$

The correct order of hydrolysis is

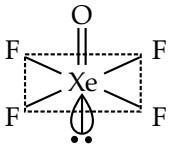
$$\underset{(p)}{CH_3CH_2\overset{\overset{O}{\|}}{C}—Cl} > \underset{(q)}{CH_3—CH_2—\overset{\overset{O}{\|}}{C}—O—\overset{\overset{O}{\|}}{C}—CH_3} > \underset{(r)}{CH_3CH_2\overset{\overset{O}{\|}}{C}—OCH_2CH_3} > \underset{(s)}{CH_3CH_2\overset{\overset{O}{\|}}{C}—NH_2}$$

45. Option (1) is correct.

A molecule with the formula AX_4Y is $XeOF_4$.

Xe is rarest monoatomic, non-radioactive element from its group.

X and Y are F and O respectively.



Square pyramidal

sp^3d^2 hydridisation

46. Correct answer is [1].

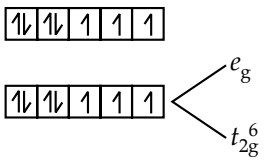
Among Mn, Cr, Co and Fe, the element with highest standard electrode potential (M^{3+}/M^{2+}) is Co its value is +1.97 V.

The number of e^- in the e_g orbital of the complex is 1.

In the complex $[CO(CN)_6]^{4-}$, the oxidation state of the cobalt is +2.

CO (Z = 27) : $3d^74s^2$

CO^{2+} : $3d^74s^0$



$\overset{\ominus}{C}N$ is a strong ligand which pair the unpaired e^-.

47. Correct answer is [6].

For the cell

$QH_2 + 2Ag^+ \rightarrow 2Ag + Q + 2H^+$

$$E = E^\circ - \frac{0.06}{2}\log\frac{[H^+]^2}{[Ag+]^2}$$

$[Ag^+] = 1\ M$

$$= E^\circ - \frac{0.06}{2}\times 2\log[H^+]$$

$$E = E^\circ - 0.06\log[H^+]$$

Here, $E = 0.4\ V$

$E^\circ_{Cell} = E^\circ_C - E^\circ_A = 0.8 - 0.7 = 0.1\ V$

$0.4 = 0.1 - 0.016\ \log[H^+]$

$$-\log[H^+] = \frac{0.4-0.1}{0.06} = 5$$

$$pH = -\log[H^+] = 5$$

From hydrolysis of SAWB

$$pH = \frac{pK_w - pK_b - \log C}{2}$$

$$5 = \frac{14 - pK_b - \log 10^{-2}}{2}$$

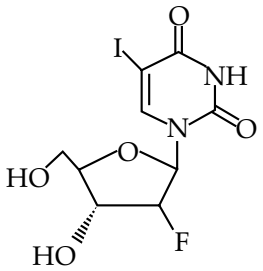
$10 = 14 - pK_b + 2$

$10 = 16 - pK_b$

$pK_b = 6$

48. Correct answer is [372].

The molecular formula of the given molecule



is $C_9N_2O_5FIH_{10}$

C = 9 × 12 = 108
N = 2 × 14 = 28
O = 5 × 16 = 80
F = 1 × 19 = 19
I = 1 × 127 = 127
H = 1 × 1 = 10
372

Mass of 1 mole = 372 g

So, 0.1 mole = 37.2 g

$= 372 \times 10^{-1}$ g

49. Correct answer is [74].

	CO	+	$2H_2$	⇌	CH_3OH
$t = 0$ (mole)	0.1		a		
t_{eq}	0.1–x		$a - 2x$		$x = 0.04$
	0.1 – 0.04		a – 0.08		

$= 0.06 = 0.023 - 0.08 = 0.15$

From $PV = nRT$

$P = 5$ bar; $V = 2$ L; $T = 500$ K; $n_{total} = 0.25$ mol

$$P_{total} = n_{total}\times\frac{RT}{V}$$

$$S = (0.06 + a - 0.08 + 0.04) \times \frac{0.08\times 500}{2}$$

$10 = (0.02 + a) \times 0.08 \times 500$

$a = 0.25 - 0.02 = 0.23$ mol

$$Kp = \frac{X_{CH_3OH}}{(X_{CO})\times(X_{H_2})}\times\frac{1}{(P_T)^2}$$

$$= \frac{0.04}{0.06\times(0.15)^2}\times\left[\frac{\frac{1}{4}}{5}\right]^2$$

$= 0.074 = 74 \times 10^{-3}$

50. Correct answer is [2435].

From the graph, it is clear that it is zero-order reaction.

$$t_{1/2} = \frac{a_0}{2K}$$

where, $\text{slope} = \frac{1}{2K} = 76.92$

$$K = \frac{1}{2\times 76.92}$$

$$[A]_t = [A_0] - Kt$$

$$[A]_{10} = 2.5 - \frac{1}{2\times 76.92}\times 10$$

$= 2.435$ mol/L

$= 2435 \times 10^{-3}$ mol/L

Mathematics

Section A

51. Option (2) is correct.

$$n = \left[\frac{50}{3}\right]+\left[\frac{50}{3^2}\right]+\left[\frac{50}{3^3}\right]+\left[\frac{50}{3^4}\right]\ldots\ldots$$

{[ ] = Greatest integer function}

$$n = 16 + 5 + 1 + 0 \ldots\ldots$$

$$= 22$$

Maximum value of $n = 22$

52. Option (3) is correct.

$$\frac{x^2}{a^2}-\frac{y^2}{b^2} = 1$$

$$\text{focus} = (\sqrt{10},0)$$

$$\text{Transverse axis} = 2a$$

$$\text{Minor axis} = 2b$$

$$\text{Eccentricity} = e$$

$$\text{Directrix} \Rightarrow x = \frac{a}{e}$$

$$ae = \sqrt{10} \text{ and } \frac{a}{e} = \frac{9}{\sqrt{10}}$$

$$\Rightarrow \quad a^2 = 9 \text{ and } e = \frac{\sqrt{10}}{3}$$

Now, $(ae)^2 = a^2 + b^2$

$$10 = 9 + b^2$$

$$\therefore \quad b = 1$$

$$\text{latus rectum } (l) = \frac{2b^2}{a} = \frac{2\times1}{3}$$

$$\Rightarrow 9(e^2 + 1)$$

$$= 9\left(\frac{10}{9}+\frac{2}{3}\right)$$

$$= 10 + 6$$

$$= 16$$

53. Option (3) is correct.

Sequence contains five 1s, three 2s, two 0s.

$\therefore$ Total number of sequences

$$= \frac{10!}{5!3!2!}$$

$$= 2{,}520$$

54. Option (2) is correct.

$$(\sin x \cos y)(f(2x + 2y)) - f(2x - 2y) = (\cos x \sin y)$$

$$(f(2x + 2y) + f(2x - 2y))$$

$$f(2x + 2y)\cdot(\sin x\cos y - \cos x \sin y)$$

$$= f(2x - 2y)\cdot(\sin x \cos y + \cos x \sin y)$$

$$f(2x + 2y)\cdot\sin(x - y) = f(2x - 2y)\cdot\sin(x + y)$$

$$\frac{f(2x+2y)}{\sin(x+y)} = \frac{f(2x-2y)}{\sin(x-y)}$$

Put $2x + 2y = m, 2x - 2y = n$

$$\frac{f(m)}{\sin\left(\frac{m}{2}\right)} = \frac{f(n)}{\sin\left(\frac{n}{2}\right)} = \text{K}$$

$$\therefore \quad \frac{f(m)}{\sin\left(\frac{m}{2}\right)} = \text{K}$$

$$f(m) = \text{K}\sin\left(\frac{m}{2}\right)$$

$$f(x) = \text{K}\sin\left(\frac{x}{2}\right)$$

$$f'(x) = \frac{\text{K}}{2}\cos\left(\frac{x}{2}\right)$$

Put $x = 0; \frac{1}{2} = \frac{\text{K}}{2} \Rightarrow \text{K} = 1$

$$f'(x) = \frac{1}{2}\cos\frac{x}{2}$$

$$f''(x) = -\frac{1}{4}\sin\frac{x}{2}$$

$$24f''\left(\frac{5\pi}{3}\right) = \left(-\frac{1}{4}\sin\left(\frac{5\pi}{6}\right)\right)24$$

$$= -\frac{24}{4}\sin\left(\frac{5\pi}{6}\right)$$

$$= -6\sin\left(\pi-\frac{\pi}{6}\right)$$

$$= -6\sin\frac{\pi}{6} = -6\times\frac{1}{2} = -3$$

55. Option (4) is correct.

$$A = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix},\ \alpha > 0,$$

$$\det(A) = 0 \Rightarrow |A| = 0$$

$$\begin{vmatrix} \alpha & -1 \\ 6 & \beta \end{vmatrix} = 0$$

$$\alpha\beta + 6 = 0$$

$$\alpha\beta = -6$$

Given, $\alpha + \beta = 1$

$$\therefore \quad \alpha + \left(\frac{-6}{\alpha}\right) = 1$$

$$\alpha^2 - 6 = \alpha$$

$$\alpha^2 - \alpha - 6 = 0$$

$$\alpha = 3, -2$$

But, $\alpha > 0 \Rightarrow \alpha = 3$

$$\beta = -2$$

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

$$\therefore \quad A^2 = A$$

$$A = A^2 = A^3 = A^4 = A^5$$

$$(I + A)^8 = I + {}^8C_1A^7 + {}^8C_2A^6 + \ldots + {}^8C_8A^8$$

$$= I + A({}^8C_1 + {}^8C_2 + \ldots + {}^8C_8)$$

$$= I + A(2^8 - 1)$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 765 & -255 \\ 1530 & -510 \end{bmatrix}$$

$$= \begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$$

56. Option (1) is correct.

$$\left(\frac{(x+1)}{(x^{2/3} + 1 - x^{1/3})} - \frac{(x+1)}{(x - x^{1/2})}\right)^{10},\ x > 1$$

$$= \left[\frac{(x^{\frac{1}{3}})^3 + 1^3}{x^{\frac{2}{3}} + 1 - x^{\frac{1}{3}}} - \frac{(\sqrt{x})^2 - 1^2}{\sqrt{x}(\sqrt{x} - 1)}\right]^{10}$$

We know that, $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$

$$\therefore \quad (x^{\frac{1}{3}})^3 + 1^3 = (x^{\frac{1}{3}} + 1)(x^{\frac{2}{3}} + 1 - x^{\frac{1}{3}})$$

$$(\sqrt{x})^2 - 1 = (\sqrt{x} + 1)(\sqrt{x} - 1)$$

$$\Rightarrow \left(\frac{(x+1)}{x^{\frac{2}{3}} + 1 - x^{\frac{1}{3}}} - \frac{(x-1)}{\left(x - x^{\frac{1}{2}}\right)}\right)^{10}$$

$$= \left((x^{\frac{1}{3}} + 1) - \left(\frac{\sqrt{x} + 1}{\sqrt{x}}\right)\right)^{10}$$

$$= \left(x^{\frac{1}{3}} - \frac{1}{\sqrt{x}}\right)^{10}$$

$$T_{r+1} = {}^{10}C_r (x)^{\frac{10-r}{3}} (-1)^r (x)^{-\frac{r}{2}}$$

$$= {}^{10}C_r (-1)^r (x)^{\frac{10-r}{3} - \frac{r}{2}}$$

Term independent of $x = \frac{10-r}{3} - \frac{r}{2} = 0$

$$(20 - 2r) - 3r = 0$$

$$r = 4$$

$$\Rightarrow \quad {}^{10}C_4(-1)^4 = 210$$

57. Option (3) is correct.

$$2\sqrt{2}\cos^2\theta + 2\cos\theta - \sqrt{6}\cos\theta - \sqrt{3} = 0$$

$$(2\cos\theta - \sqrt{3})(\sqrt{2}\cos\theta + 1) = 0$$

$$\cos\theta = \frac{\sqrt{3}}{2}, \frac{-1}{\sqrt{2}}$$

In $[0, 2\pi]$, $\cos\theta = \frac{\sqrt{3}}{2}$, then $\theta = \pm\frac{\pi}{6}, \pm\frac{11\pi}{6}$

If $\cos\theta = \frac{-1}{\sqrt{2}}$, then $\theta = \pm\frac{3\pi}{4}, \pm\frac{5\pi}{4}$

Hence, there are a total of 8 solutions because cos$x$ is an even function.

58. Option (1) is correct.

Let first term $a_1 = a$

Common difference $= d$

$$\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$$

$$\frac{12}{2}[2a+11\times 2d] = -\frac{72}{5}a_1$$

$$12a + 132d = -\frac{72}{5}a_1$$

$$132a + 132 \times 5d = 0$$

$$a = -5d$$

If $\sum_{k=1}^{n} a_k = 0,$

$$\frac{n}{2}(2a+(n-1)d) = 0$$

$$\Rightarrow \quad -10d + nd - d = 0$$

$$n = 11$$

59. Option (4) is correct.

$$f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$$

$$f'(x) = 6x^2 - 18ax + 12a^2$$

$$f'(x) = 6(x^2 - 3ax + 2a^2)$$

$$f'(x) = 0$$

$$x^2 - 3ax + 2a^2 = 0$$

$$x = a, 2a$$

$$f''(x) = 6(2x - 3a)$$

$$f''(a) = 6(2a - 3a)$$

$$= -6a < 0$$

$\therefore$ Local maxima at $x = a$,

Local minima at $x = 2a$

Given: $p^2 = q$

$\Rightarrow \quad a^2 = 2a$

$$a(a - 2) = 0$$

$\Rightarrow \quad a = 0$ or 2

Since $a > 0$, take $a = 2$

$$f(x) = 2x^3 - 18x^2 + 48x + 1$$

$$f(3) = 37$$

60. Option (1) is correct.

$$\frac{2+k^2z}{k+\bar{z}} = kz$$

$$2 + k^2z = k^2z + kz\bar{z}$$

$$2 = kz\bar{z}$$

$$\{|z|^2 = z\bar{z}\ (\because |z| = 1)\}$$

$\Rightarrow \quad 2 = k \times 1$

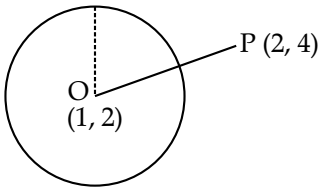
$$k + ik^2 = 2 + 4i$$

Let $P = (2, 4)$

Given, $|z - (1 + 2i)| = 1$

Centre of circle $= (1, 2)$

Radius $(r) = 1$



Maximum distance of P from circle = OP + r

$$(OP + r) = \sqrt{1+4} + 1 = \sqrt{5} + 1$$

61. Option (3) is correct.

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$$a_1^2 + a_2^2 + a_3^2 = 1$$

Let $\vec{b} = 2\vec{i} - \hat{j} + 2\hat{k},\ \vec{c} = \vec{i} + 2\hat{j} - 2\hat{k}$

$$\vec{d} = \hat{k}$$

Given projections are equal therefore,

$$\frac{\vec{a}\cdot\vec{b}}{|b|} = \frac{\vec{a}\cdot\vec{c}}{|c|} = \frac{\vec{a}\cdot\vec{d}}{|d|}$$

$$\frac{2a_1 - a_2 + 2a_3}{3} = \frac{a_1 + 2a_2 - 2a_3}{3} = a_3$$

$\therefore \quad 2a_1 - a_2 + 2a_3 = a_1 + 2a_2 - 2a_3$

$$a_1 - 3a_2 + 4a_3 = 0 \qquad ...(i)$$

$$2a_1 - a_2 + 2a_3 = 3a_3$$

$$2a_1 - a_2 - a_3 = 0 \qquad ...(ii)$$

$$a_1 + 2a_2 - 2a_3 = 3a_3$$

$$a_1 + 2a_2 - 5a_3 = 0 \qquad ...(iii)$$

From Eq. (ii), we get

$$a_2 + a_3 = 2t \qquad \{\text{Let, } a_1 = t\}$$

From Eq. (i), we get

$$3a_2 - 4a_3 = t$$

$\therefore \quad 3a_2 - 4(2t - a_2) = t$

$$3a_2 + 4a_2 - 8t = t$$

$$7a_2 = 9t$$

$$a_2 = \frac{9t}{7}$$

Now, $a_2 + a_3 = 2t$

$$\frac{9t}{7} + a_3 = 2t$$

$\therefore \quad a_3 = \frac{5t}{7}$

Hence, $a_1 : a_2 : a_3 = t : \frac{9t}{7} : \frac{5t}{7}$

$$= 7 : 9 : 5$$

$\therefore$ Unit vector along $\vec{a} = \frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} + 5\hat{k})$

62. Option (2) is correct.

$R = \{(f, g) : f(0) = g(1) \text{ and } f(1) = g(0)\}$

Reflexive: $(f, f) \in R$

$= f(0) = f(1)$ and $f(1) = f(0) \rightarrow$ must hold

$\Rightarrow$ but this is not true for all function.

So, it is not reflexive.

Symmetric: If $(f, g) \in R \Rightarrow (g, f) \in R$

Now, $g(0) = f(1)$ and $g(1) = f(0) \rightarrow$ true

$\therefore$ symmetric

Transitive: If $(f, g) \in R$ and $(g, h) \in R$

$\Rightarrow (f, h) \in R$

Now, $(f, g) \in R \Rightarrow f(0) = g(1)$ and $f(1) = g(0)$

$(g, h) \in R \Rightarrow g(0) = h(1)$ and $g(1) = h(0)$

For $(f, h) \in R$, we need $f(0) = h(1)$ and $f(1) = h(0)$

Now, $f(0) = g(1) = h(0)$ and $f(1) = g(0) = h(1)$

Hence, it is not transitive.

It is symmetric but neither reflective nor transitive.

63. Option (1) is correct.

$$\lim_{x\to 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$$

Applying following Frenies expansion

$$\sin t = t - \frac{t^3}{3!} + \ldots$$

$$e^{x^2} = 1 + \frac{x^2}{1} + \frac{x^4}{2!} \ldots.$$

Neglecting higher degree terms,

$$\lim_{x\to 10} \frac{x^2(\alpha x) + (\gamma - 1)\left(1 + \frac{x^2}{1}\right)}{2x - \frac{8x^3}{6} - \beta x} = 3$$

$$\lim_{x\to 0} \frac{(\gamma - 1) + (\gamma - 1)x^2 + \alpha x^3}{(2 - \beta)x - \frac{4}{3}x^3}$$

Limit will exist if coefficients of x^0, x^1, x^2 are zero.

$$\gamma - 1 = 0, 2 - \beta = 0$$

and $$\frac{\alpha}{\left(\frac{-4}{3}\right)} = 3$$

$$\gamma = 1, \beta = 2, \frac{-3\alpha}{4} = 3$$

$\Rightarrow$ $$\alpha = -4$$

$$\beta + \gamma - \alpha = 7$$

64. Option (2) is correct.

$$3x + y + \beta z = 3$$

$$2x + \alpha y - z = -3$$

$$x + 2y + z = 4$$

$$D = \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$D_x = \begin{vmatrix} 3 & 1 & \beta \\ -3 & \alpha & -1 \\ 4 & 2 & 1 \end{vmatrix}$$

$$D_y = \begin{vmatrix} 3 & 3 & \beta \\ 2 & -3 & -1 \\ 1 & 4 & 1 \end{vmatrix}$$

$$D_z = \begin{vmatrix} 3 & 1 & 3 \\ 2 & \alpha & -3 \\ 1 & 2 & 4 \end{vmatrix}$$

There are infinite many solution,

therefore, $D = 0 = D_x = D_y = D_z$

$$D = 0 = \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$3\alpha + 4\beta - \alpha\beta + 3 = 0$$

$$D_z = 0 = \begin{vmatrix} 3 & 1 & 3 \\ 2 & \alpha & -3 \\ 1 & 2 & 4 \end{vmatrix}$$

$$0 = 9\alpha + 19$$

$$\alpha = \frac{-19}{9}, \beta = \frac{6}{11}$$

$\Rightarrow$ $$22\beta - 9\alpha = 31$$

65. Option (2) is correct.

$$P_n = \alpha^n + \beta^n$$

$$P_{10} = \alpha^{10} + \beta^{10} = 123$$

$$P_9 = \alpha^9 + \beta^9 = 76$$

$$P_8 = \alpha^8 + \beta^9 = 47$$

$$P_1 = \alpha + \beta = 1$$

$$P_9 + P_8 = \alpha^9 + \beta^9 + \alpha^8 + \beta^8$$

$$= 76 + 47$$

$$\alpha^8(\alpha + 1) + \beta^8(\beta + 1) = 123$$

$\therefore$ $$\alpha^8(\alpha + 1) + \beta^8(\beta + 1) = \alpha^{10} + \beta^{10}$$

$\Rightarrow$ $$\alpha^2 = \alpha + 1$$

and $$\beta^2 = \beta + 1$$

Given quadratic equation is,

$$x^2 = x + 1$$
$$x^2 - x - 1 = 0$$
$$\alpha + \beta = 1,\ \alpha\beta = -1$$
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{1}{-1}$$
$$\frac{1}{\alpha}\cdot\frac{1}{\beta} = \frac{1}{\alpha\beta} = -1$$

$\therefore$ Quadratic equation is

$$x^2 - x(-1) - 1 = 0$$
$$x^2 + x - 1 = 0$$

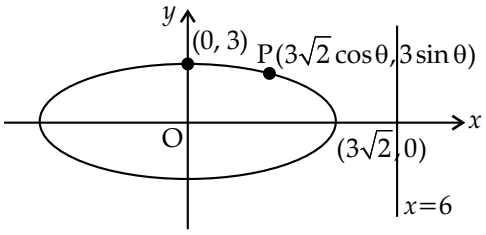
66. Option (4) is correct.

$$\frac{x^2}{18} + \frac{y^2}{9} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

On comparison, we get

$$a^2 = 18,\ b^2 = 9 \Rightarrow a > b$$



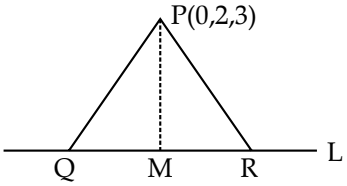
$$PS + PS' = 2\times 3\sqrt{2}$$
$$b^2 = a^2(1 - e^2) = 18(1 - e^2)$$
$$\Rightarrow \quad e = \frac{1}{\sqrt{2}}$$

Directrix $\quad x = \dfrac{3\sqrt{2}}{\frac{1}{\sqrt{2}}} = 6$

$$SP = a(1 - e\cos\theta)$$
$$S'P = a(1 + e\cos\theta)$$
$$SP\cdot S'P = a^2(1 - e\cos\theta)(1 + e\cos\theta)$$
$$= a^2(1 - e^2\cos^2\theta)$$
$$= 18\left(1 - \frac{\cos^2\theta}{2}\right)$$

$$(SP\cdot S'P)_{max} = 18 \text{ if } \cos\theta = 0$$
$$(SP\cdot S'P)_{min} = 9 \quad \text{if } \cos\theta = 1$$
$$\text{Sum} = 18 + 9 = 27$$

67. Option (2) is correct.



$$QR = 5$$

$\therefore \quad \dfrac{x+3}{5} = \dfrac{y-1}{2} = \dfrac{z+4}{3} = \lambda$

Any point on line L is: $5\lambda - 3, 2\lambda + 1, 3\lambda - 4$

Let, $M = 5\lambda - 3, 2\lambda + 1, 3\lambda - 4$

Drs of PM $\Rightarrow 5\lambda - 3, 2\lambda - 1, 3\lambda - 7$

Drs of line L $\Rightarrow 5, 2, 3$

L and PM are perpendicular,

$\therefore \quad 5(5\lambda - 3) + 2(2\lambda - 1) + 3(3\lambda - 7) = 0$

$$25\lambda - 15 + 4\lambda - 2 + 9\lambda - 21 = 0$$
$$38\lambda - 38 = 0$$
$$\lambda = 1$$

$\therefore \quad M = (2, 3, -1)$

$$PM = \sqrt{4+1+16} = \sqrt{21}$$
$$\text{Area} = \frac{1}{2}\times 5\times\sqrt{21} = \frac{m}{n}$$
$$2m - 5\sqrt{21}n = 0$$

68. Option (3) is correct.

Edges AB, AC and AD are mutually perpendicular,

$\therefore$ Ar(ΔABCD)

$$= \sqrt{(\text{Ar}(\Delta ABC))^2 + (\text{Ar}(\Delta ACD))^2 + (\text{Ar}(\Delta ADB))^2}$$
$$= \sqrt{5^2 + 6^2 + 7^2}$$
$$= \sqrt{110}$$

69. Option (4) is correct.

$$A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$$

$\therefore \quad A = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix} - I = \begin{bmatrix} 0 & a & 1 \\ 2 & 0 & 0 \\ a & 1 & 1 \end{bmatrix}$

$$|A| = \begin{vmatrix} 0 & a & 1 \\ 2 & 0 & 0 \\ a & 1 & 1 \end{vmatrix}$$

$$|A| = 2 - 2a$$

Given, $\quad |A| = -4$

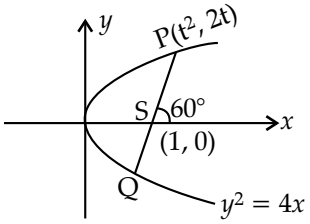
$\therefore \quad 2 - 2a = -4$

$$a = 3$$

$|(a + 1)\text{adj}(a - 1)A| = |4 \text{ adj } 3A|$

$$= 4^3|\text{adj } 3A|$$
$$= 4^3 \times |3A|^{3-1} = 64|3A|^2$$
$$= 64 \times (3^3)^2 |A|^2$$
$$= 2^6 \times 3^6 \times 16 \quad (\because |A| = -4)$$
$$2^m \times 3^n = 2^{10} \times 3^6$$
$$\therefore \quad m = 10, n = 6$$
$$\therefore \quad m + n = 16$$

70. Option (1) is correct.



Parabola $\Rightarrow y^2 = 4x$

Let, parametric coordinate of P $= (t^2, 2t)$

Focus (S) $= (1, 0)$

$$\tan 60° = \frac{2t-0}{t^2-1} = \sqrt{3} \Rightarrow t = \sqrt{3}$$

$\therefore$ P$(3, 2\sqrt{3})$

Equation of circle in diameter for m is:

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$(x - 1)(x - 3) + (y - 0)(y - 2\sqrt{3}) = 0$

at $x = 0$

$\Rightarrow \quad 3 + y^2 - 2\sqrt{3}y = 0$

$\Rightarrow \quad y = \sqrt{3} = \alpha$

$5\alpha^2 = 15$

Section B

71. Correct answer is [8].

Let $f(x) = \dfrac{1}{e^{x-1}}$

$f(x) = e^{1-x}$

If, $f(x) = 1 \Rightarrow 1 - x = 0$

$\therefore \quad x = 1$

$f(x) = 2$

$\dfrac{1}{e^{x-1}} = 2$

$x = 1 - \ln 2$

$f(0) = e^1 = 2.71$

$f(e^3) = e^{1-e^3} \in (0, 1)$

$$I = \int_0^{e^3} [e^{1-x}]dx = \int_0^{1-\ln 2} [e^{1-x}]dx + \int_{1-\ln 2}^{1} [e^{1-x}]dx + \int_1^{e^3} [e^{1-x}]dx$$

$$I = \int_0^{1-\ln 2} 2dx + \int_{1-\ln 2}^{1} 1dx + \int_1^{e^3} 0dx$$

$$= 2(1 - \ln 2 - 0) + 1(1 - 1 + \ln 2) + 0$$

$\alpha - \ln 2 = 2 - \ln 2$

$\Rightarrow \quad \alpha = 2$

$\Rightarrow \quad \alpha^3 = 8$

72. Correct answer is [36].

$$f''(x) = f(x)$$

Multiplying $f'(x)$ on both sides, we get

$$f''(x) f'(x) = f(x)f'(x)$$

On integration both sides, we get

$$\int f''(x)f'(x)dx = \int f(x)f'(x)dx$$

$$\Rightarrow \quad \frac{(f'(x))^2}{2} = \frac{(f(x))^2}{2} + C$$

C is the constant of integration,

Take $x = 0$

$$\frac{(f'(0))^2}{2} = \frac{(f(0))^2}{2} + C$$

$$\frac{9}{2} = C$$

$$(f'(x))^2 = (f(x))^2 + 9$$

$$f'(x) = \sqrt{(f(x))^2 + 9} \quad \because f'(x) \geq 0$$

$$\int dx \Rightarrow \ln y + \sqrt{y^2+9} = x + C \qquad \{\text{Let } f(x) = y\}$$

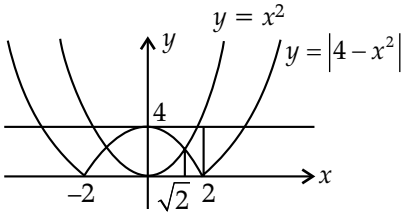
$$\Rightarrow \quad f(0) = 0 \Rightarrow C = \ln 3$$

$$\Rightarrow \quad y + \sqrt{y^2+9} = 3e^x$$

at $x = \ln 3; y = 4$

$\therefore \quad 9f(\ln 3) = 36$

73. Correct answer is [22].



$$A = \int_0^4 \sqrt{4+y}dy - \int_0^2 \sqrt{4-y}dy - \int_0^4 \sqrt{y}dy$$

$$= \left(\frac{(4+y)^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^4 + \left(\frac{(4-y)^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^2 - \left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^4$$

$$= \frac{40\sqrt{2}}{3} - 16 = \frac{80\sqrt{2}}{6} - 16$$

$$A = \frac{80\sqrt{2}}{\alpha} - \beta$$

$\therefore \quad \alpha = 6, \beta = 16$

$\alpha + \beta = 22$

74. Correct answer is [2477].

$1 \le a < ar < ar^2 \le 40$

Let first term of GP $= a$

Common ratio $= r$

Case I: If r is natural number.

If $r = 2$

$1 \le a < 2a < 4a \le 40$

$a \in \{1, ..., 10\}$ (10 G.P.)

If $r = 3$

$1 \le a < 3a < 9a \le 40$

$a \in \{1, 2, 3, 4\}$ (4 G.P.)

If $r = 4$

$1 \le a < 4a < 16a \le 40$

$a \in \{1, 2\}$ (2 GP)

If $r = 5$

$1 \le a < 5a < 25a \le 40$

$a \in \{1\}$ (1 GP)

If $r = 6$

$1 \le a < 6a < 36a \le 40$

$a \in \{1\}$ (1 GP)

Case II: If r is fraction > 1

If $r = \frac{3}{2}$

then $a, \frac{3a}{2}, \frac{9a}{4}$

A must be multiple of 4

$a \in \{4, 8, 12, 16\}$

$\left.\begin{matrix}(4,6,9)\\(8,12,18)\\(12,18,27)\\(16,24,36)\end{matrix}\right\}$ 4, GP

If $r = \frac{5}{2}$, then $a, \frac{5a}{2}, \frac{25a}{4}$

$a = 4$ ...(1) GP

$r = \frac{4}{3}, \quad ar^2 = \frac{16a}{9} \rightarrow a = 9k$

(9, 12, 16), (18, 24, 32) ...(2) GP

$r = \frac{5}{3}, \quad ar^2 = \frac{25a}{9}; a = 9k$

(9, 15, 25) ...(1) GP

$r = \frac{5}{4} \quad ar^2 = \frac{25a}{16}; a = 16k$

(16, 20, 25) ...(1) GP

$r = \frac{6}{5} \quad ar^2 = \frac{36a}{25}; a = 25k$

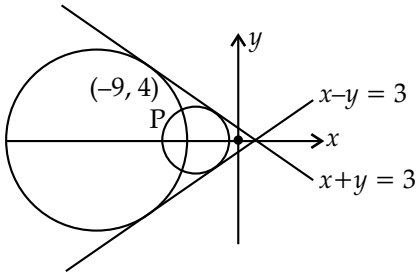
(25, 30, 36) ...(1) GP

Total $= 18 + 10 = 28$

$$P = \frac{28}{{}^{40}C_3} = \frac{28}{9880} = \frac{7}{2470}$$

$m + n = 2477$

75. Correct answer is [768].



Let centre of circle $= (a, 0)$

radius $= r$

Distance of centre from line

$x - y - 3 = 0$ is r

$\therefore \quad r = \left|\frac{a-3}{\sqrt{2}}\right|$

Equation of circle,

$$(x-a)^2 + y^2 = \left(\frac{a-3}{\sqrt{2}}\right)^2$$

$\because$ It passes through (–9, 4)

$2(a^2 + 18a + 81 + 16) = (a^2 - 6a + 9)$

$a^2 + 42a + 185 = 0$

$(a + 37)(a + 5) = 0$

$\Rightarrow \quad a = -37, -5$

As, $r = \left|\frac{a-3}{\sqrt{2}}\right|$

$$r_1 = \left|\frac{-37-3}{\sqrt{2}}\right| = 20\sqrt{2}$$

$$r_2 = \left|\frac{-5-3}{\sqrt{2}}\right| = 4\sqrt{2}$$

$|r_1^2 - r_2^2| = |800 - 32| = 768$

■■■